

GAIRA V7 — Phase 06.5

Latent Spectral Geometry Audit

What biochemical organisation emerges from the frozen CSM representation, independent of the curated ontology?

Status	COMPLETE — AUDIT ONLY, no architecture changed
Recommendation	Option A — current architecture retained
Unit of analysis	canonical molecule (154), not spectrum
Labels in construction	NONE — chemistry is an external validation target only
Preferred cluster count	NONE — 0 of 7 indices have an interior optimum
Chemistry vs acquisition	PERMANOVA R^2 0.452 vs excitation 0.118, source 0.040
Graph modularity	0.718 vs null 0.347 ($z = 40$)
Shape of the space	modular — hierarchy with continuous branch lengths
Coordinates vs hard ids	k-NN preservation 0.446 vs 0.237
Coordinate reproducibility	0.963
Coordinate robustness	mean cosine 0.958
Retrieval gain from coordinates	molecule $\Delta+0.016$ ($p = 0.180$) · chemistry $\Delta+0.003$ ($p = 1.000$) — NEITHER SIGNIFICANT
Agreement with the ontology	AMI 0.702 at $K = 16$ — but AMI is monotone in K and peaks at $K = 24$
Frozen inputs	LSM 208482d6... · CSM 0b4aa550... · engine 20d8bd99... — verified
Phase 07	NOT BEGUN
Completed	2026-08-07 02:45:45 UTC

Sources of record: reports/PHASE_06_5_LATENT_GEOMETRY_AUDIT.md and reports/PHASE_06_5_SCIENTIFIC_AUDIT.md
The Continuous Spectral Coordinates are retained as a SCIENTIFIC INSTRUMENT. They are not a chemistry probability, not a concentration, and they do not enter the GAIRA inference path.

Figures

01 stability curves

02 k16 detail

03 cluster composition

04 cluster spectra

05 activation heatmap

06 embeddings

07 knn graph

08 confounding

09 hierarchy

10 coordinates

11 coordinate robustness

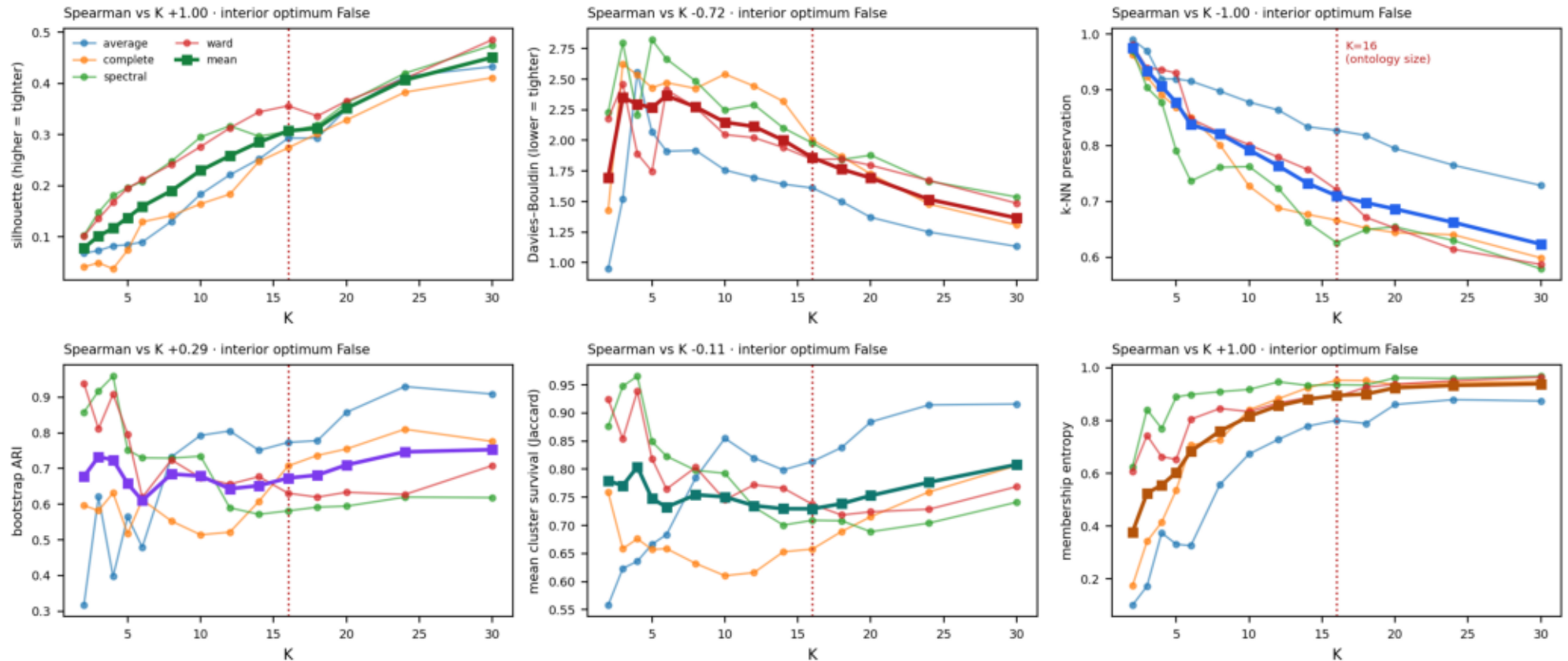
12 retrieval

13 agreement

14 decision

Figure 01

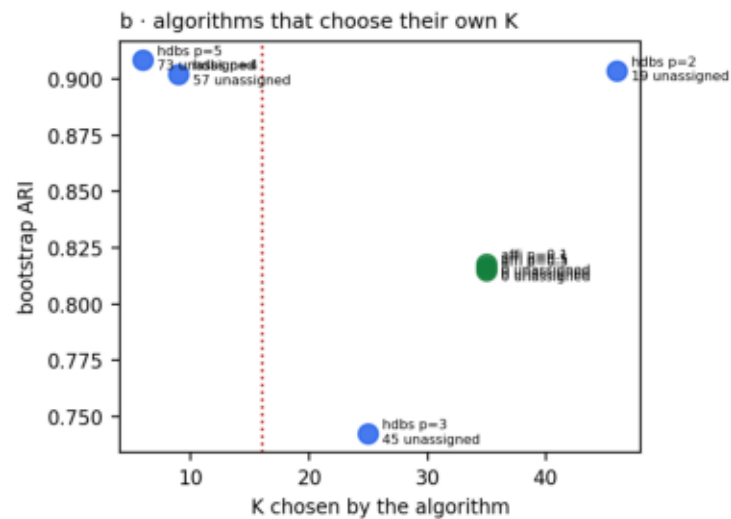
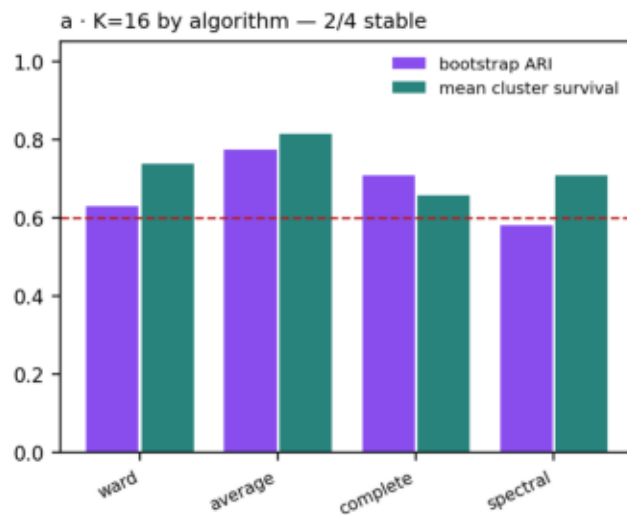
**Figure 1 · Cluster stability across 14 K and 4 fixed-K algorithms — NO index has an interior optimum
every curve is monotone or flat in K: the geometry has no preferred cluster count, which is what a continuum looks like**



Cluster stability across 14 values of K and four fixed-K algorithms. NO index has an interior optimum: silhouette rises monotonically to K=30 (Spearman +1.00), neighbour preservation falls monotonically (-1.00), membership entropy rises monotonically (+1.00). Bootstrap stability is U-shaped because coarse partitions and near-singleton partitions are both trivially reproducible. This is what a continuum looks like.

Figure 02

Figure 2 · Is K=16 genuinely stable, or merely convenient?



NO PREFERRED CLUSTER COUNT EXISTS

0 of 7 internal indices have an interior optimum.

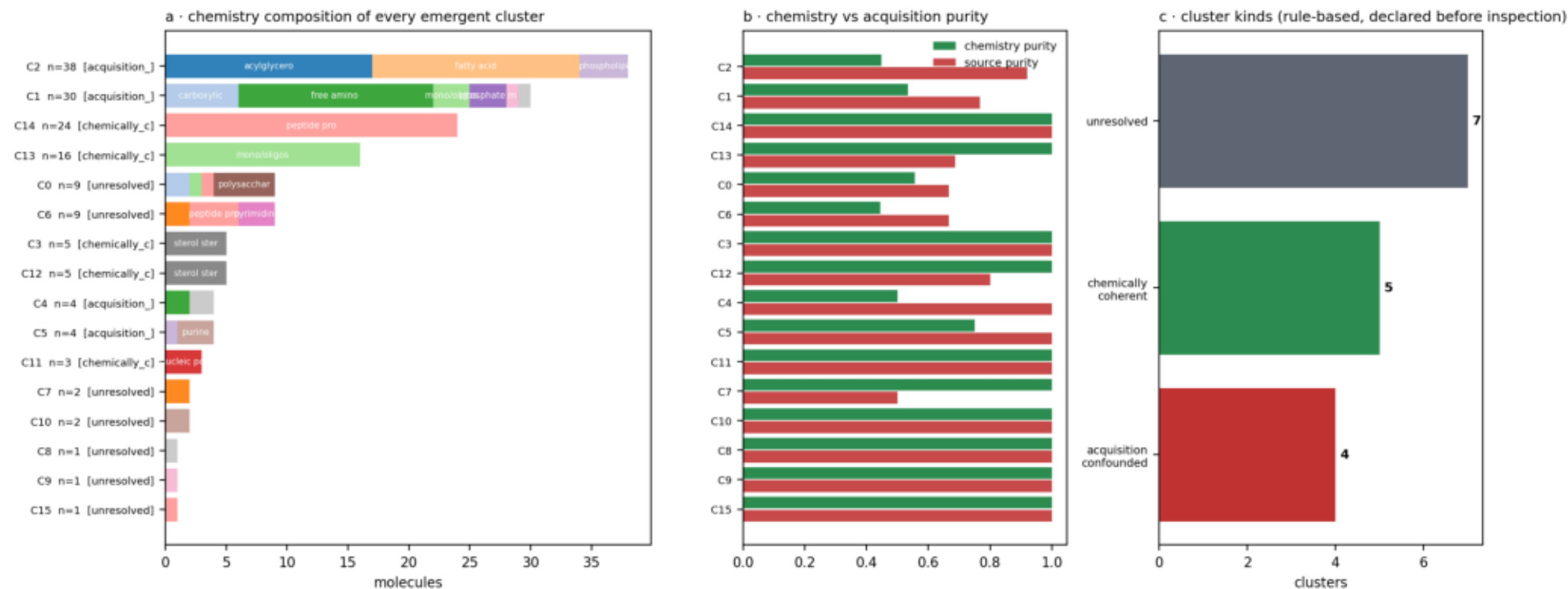
Silhouette rises monotonically to K=30 (Spearman +1.00), neighbour preservation falls monotonically to K=30 (-1.00), membership entropy rises monotonically (+1.00). Every index is tracking granularity, not structure.

K=16 is therefore adopted as a REPORTING CONVENTION — it matches the curated ontology so Section 7's comparison is like-for-like — and NOT as a discovered optimum. An earlier rule that maximised bootstrap ARI over all K chose K=4, because a coarse partition is trivially reproducible: the same stability-without-informativeness trap that principle P-18 exists to catch.

Is K=16 genuinely stable or merely convenient? Merely convenient: 2 of 4 algorithms call it stable and its mean bootstrap ARI (0.673) is unremarkable against its neighbours. The free-K algorithms choose 3-24, never 16. K=16 is adopted as a reporting convention for comparability with the ontology, not as a discovery.

Figure 03

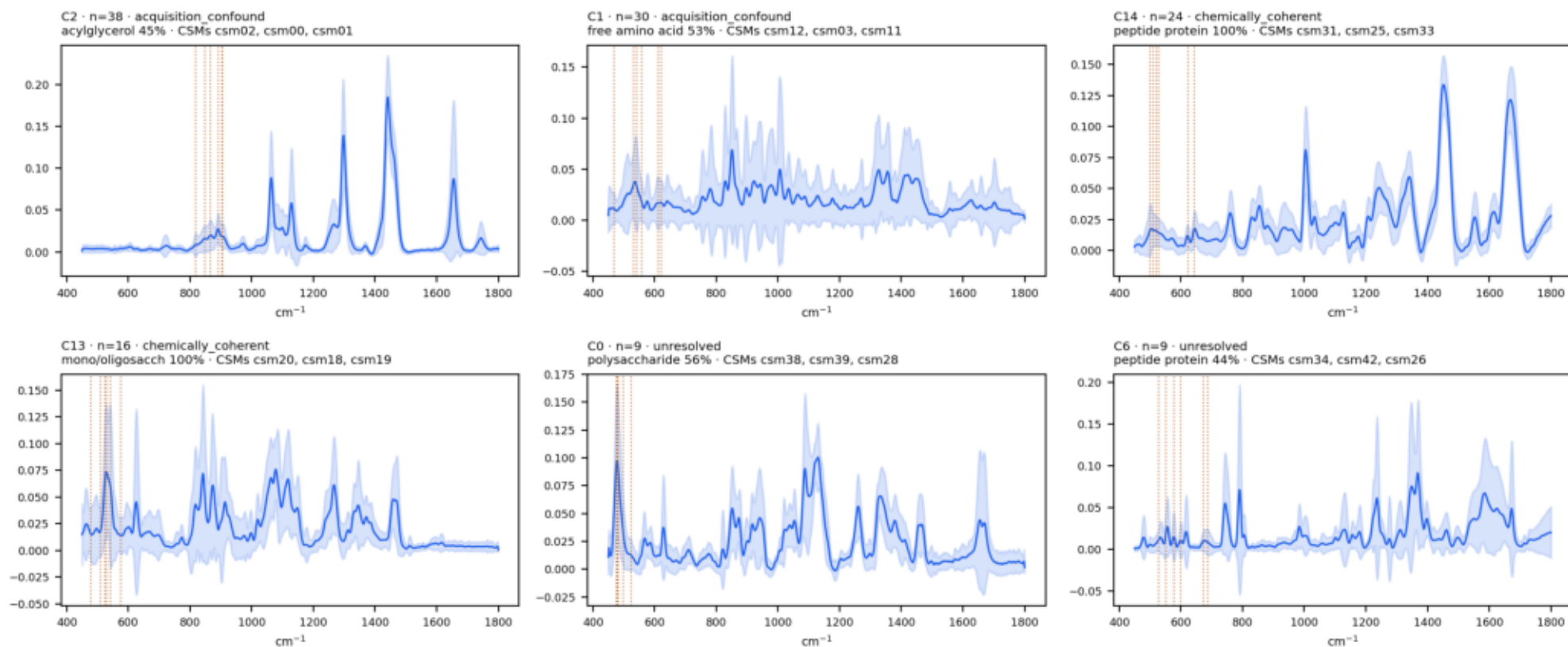
Figure 3 · What is inside each emergent cluster



Composition of every emergent cluster, and the rule-based classification whose thresholds were declared before any cluster was inspected. Five clusters are 100% chemically pure without a label having been used; four are acquisition-confounded, their source or excitation purity exceeding their chemistry purity.

Figure 04

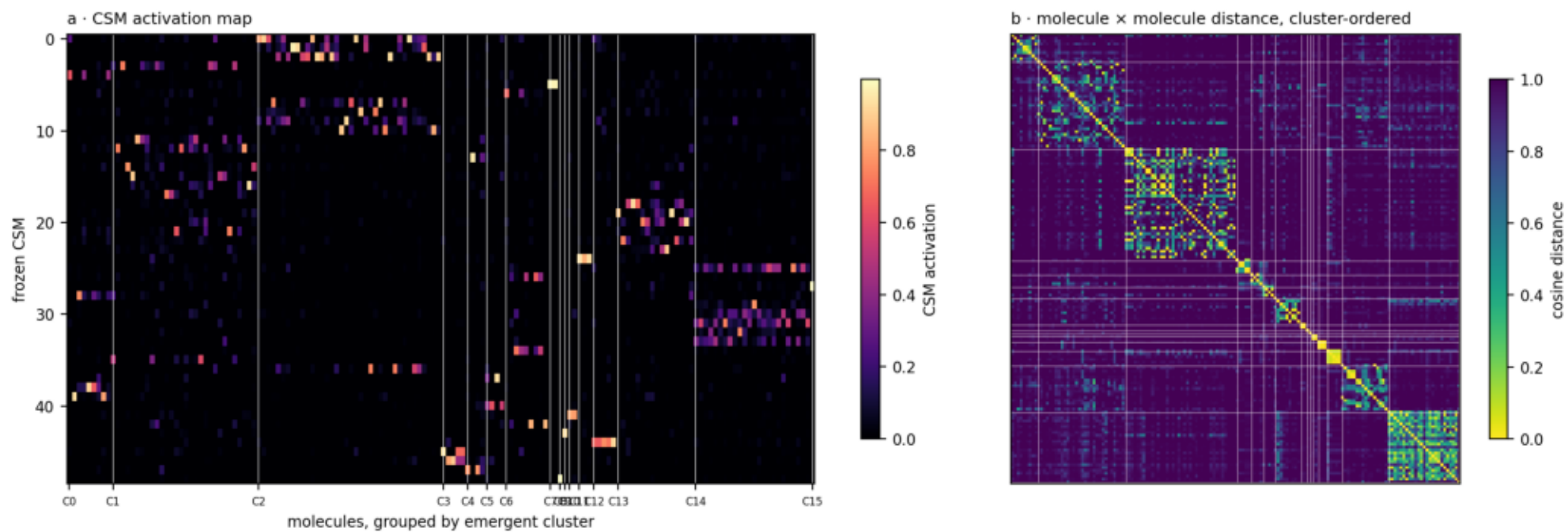
Figure 4 · Mean spectrum ± 1 sd of the six largest emergent clusters (amber = the cluster's dominant CSM bands)



Mean spectrum ± 1 sd of the six largest emergent clusters, with each cluster's dominant CSM bands marked. The clusters are spectroscopically coherent objects, not arbitrary partitions of a cloud.

Figure 05

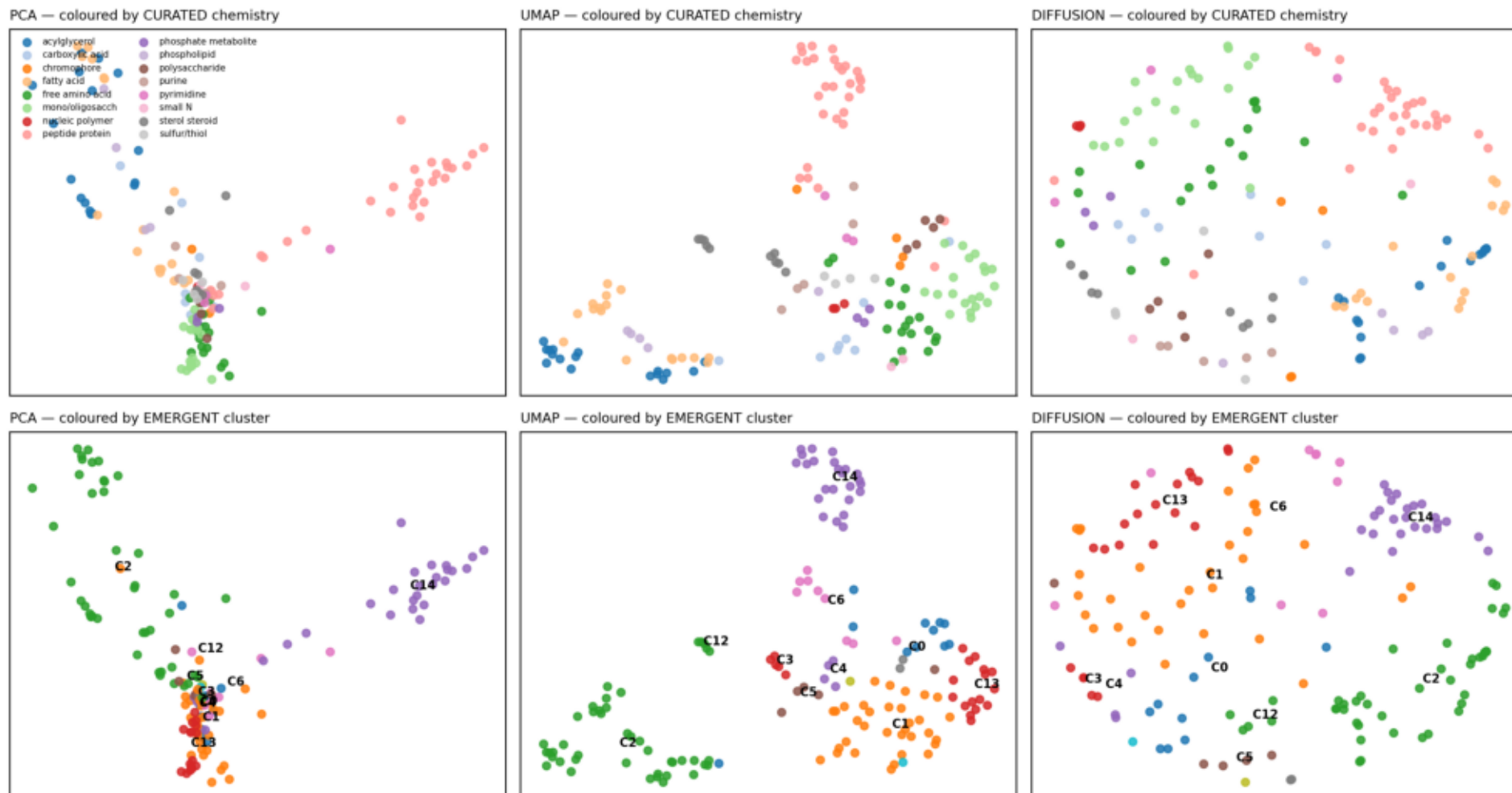
Figure 5 · The CSM manifold, ordered by the emergent partition



The CSM activation map and the molecule x molecule distance matrix, both ordered by the emergent partition. Block structure is visible and soft-edged.

Figure 06

Figure 6 · Embeddings for interpretation only — never inference
PCA components 1-2 explain 15.5% of variance; UMAP distances must not be read quantitatively



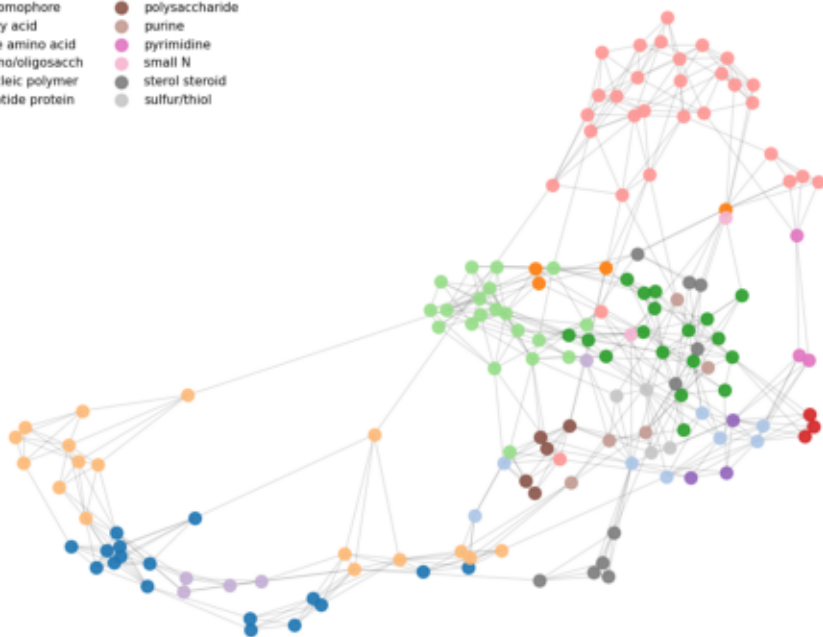
PCA, UMAP and MDS embeddings coloured by curated chemistry (top) and by emergent cluster (bottom). Visualisation only: PCA components 1-2 explain 15.5% of variance and UMAP distances must not be read quantitatively.

Figure 07

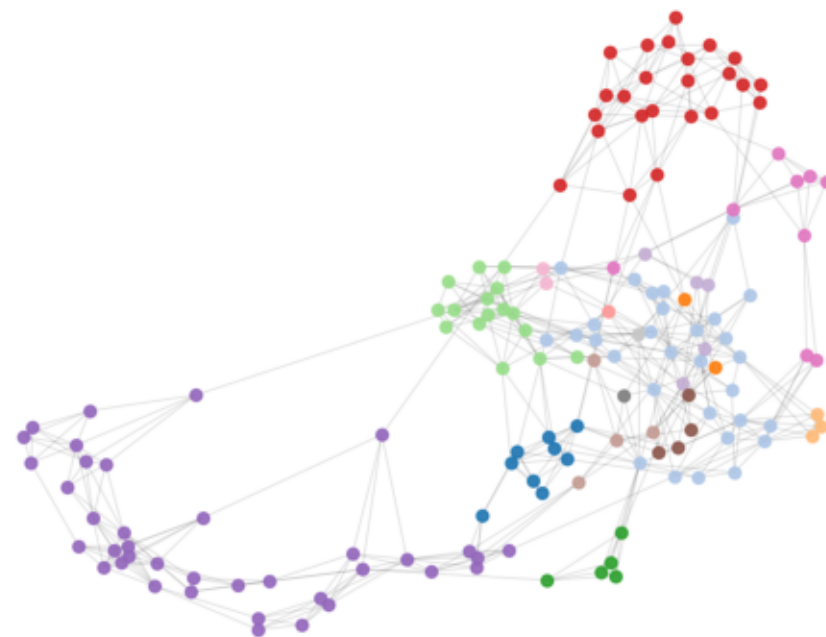
**Figure 7 • Force-directed 5-NN graph of the 154 molecules in CSM space
modularity 0.718 against a degree-preserving null of 0.347 ($z = 40$) — the graph is genuinely modular, in 9 communities**

coloured by curated chemistry

- | | |
|-------------------|------------------------|
| ● acylglycerol | ● phosphate metabolite |
| ● carboxylic acid | ● phospholipid |
| ● chromophore | ● polysaccharide |
| ● fatty acid | ● purine |
| ● free amino acid | ● pyrimidine |
| ● mono/oligosacch | ● small N |
| ● nucleic polymer | ● sterol steroid |
| ● peptide protein | ● sulfur/thiol |



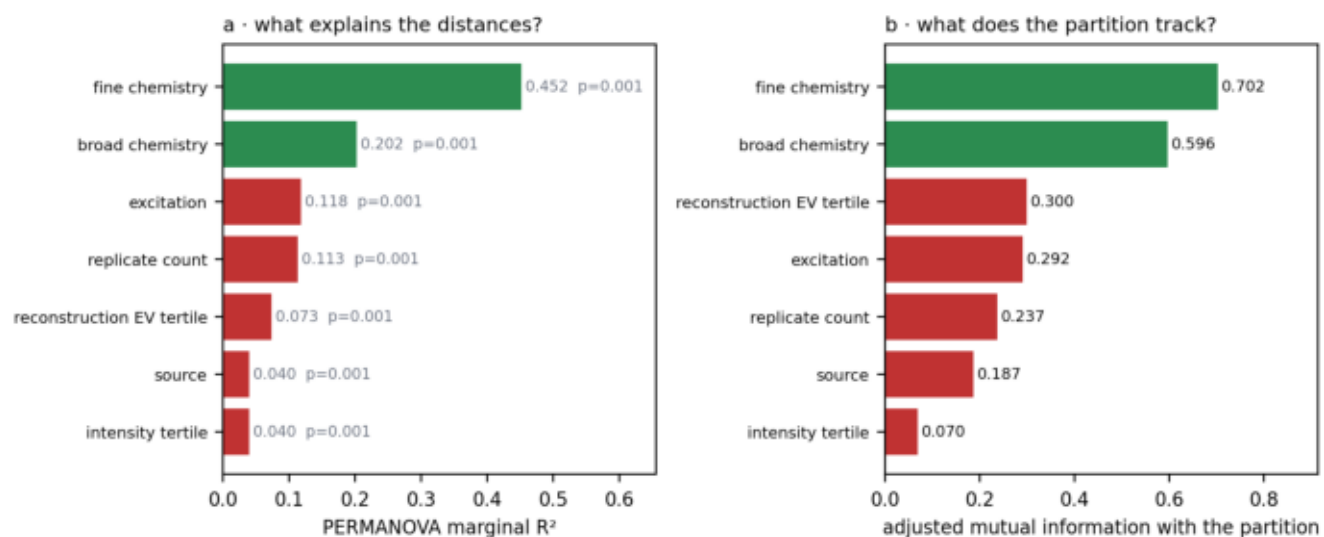
coloured by emergent cluster



Force-directed 5-NN graph of the 154 molecules. Modularity 0.718 against a degree-preserving null of 0.347 ($z = 40$) - the graph is genuinely modular - while 35% of molecules have neighbours in more than one community.

Figure 08

Figure 8 · Is the geometry chemistry, or the instrument?



CHEMISTRY DOMINATES: True

fine chemistry R^2 0.452
excitation R^2 0.118
source R^2 0.040

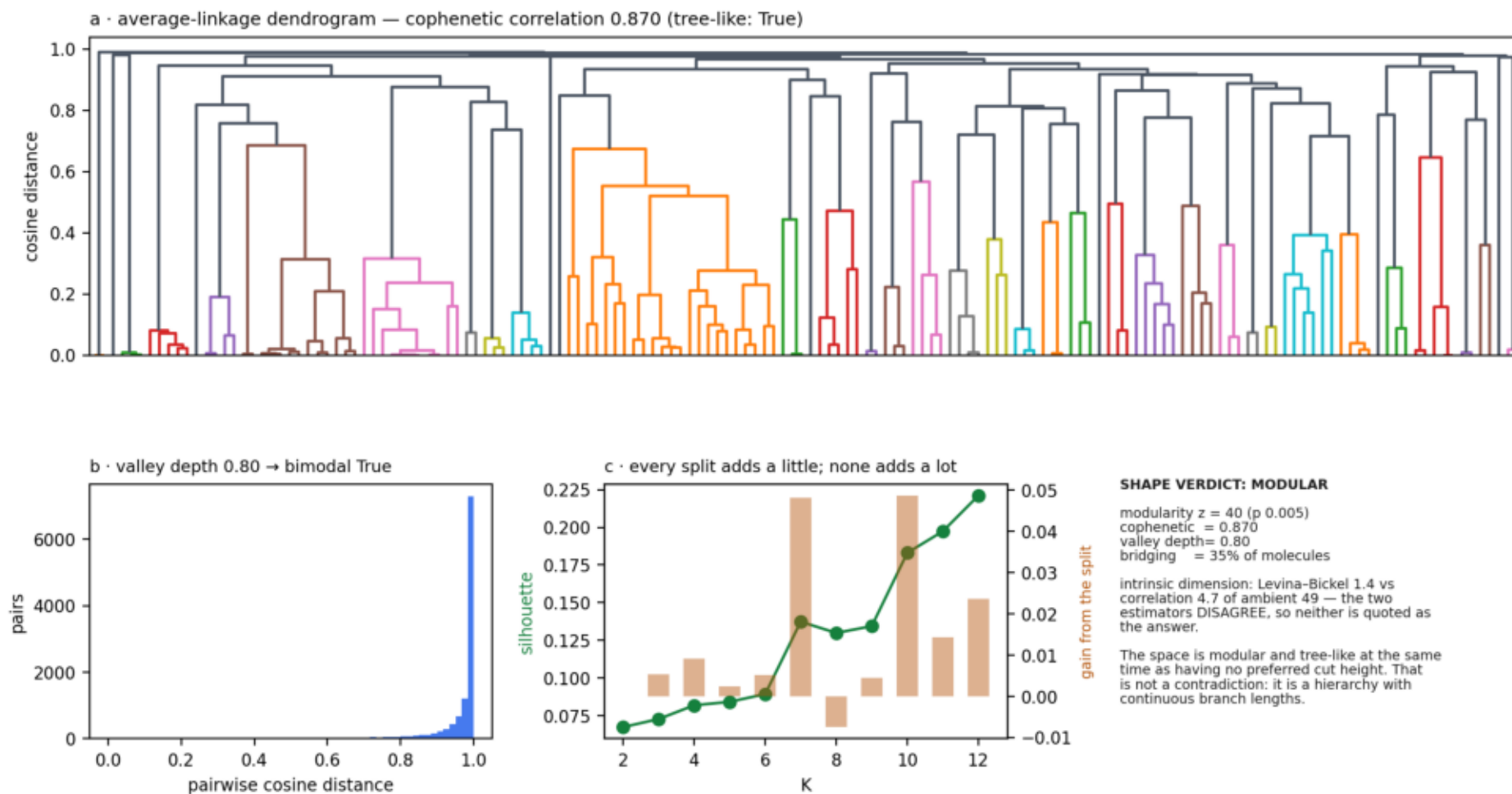
Chemistry explains 3.8x more of the pairwise distance structure than excitation and 11x more than source library. All factors are individually significant at $p = 0.001$, which at $n = 154$ means detectable, not dominant.

The caveat that matters: globally chemistry wins, but FOUR of sixteen clusters are individually acquisition-confounded by the pre-declared rule — their source or excitation purity exceeds their chemistry purity. A global verdict does not license a per-cluster claim.

Is the geometry chemistry or the instrument? Chemistry explains 3.8x more of the distance structure than excitation and 11x more than source library. The caveat: globally chemistry wins, but four individual clusters are acquisition-confounded.

Figure 09

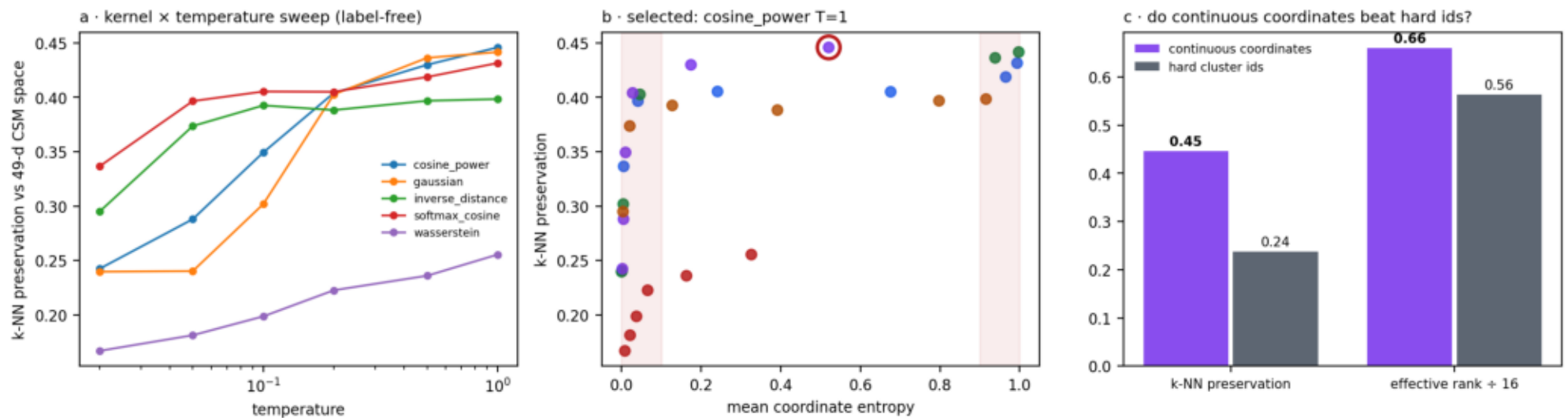
Figure 9 · Is Raman molecule space tree-like, graph-like, continuous or modular?



Dendrogram, distance distribution, split gains and the shape verdict. The space is modular ($z = 40$) and tree-like (cophenetic 0.870) while having no preferred cut height - a hierarchy with continuous branch lengths. The two intrinsic-dimension estimators disagree by 3.3x, so neither is quoted.

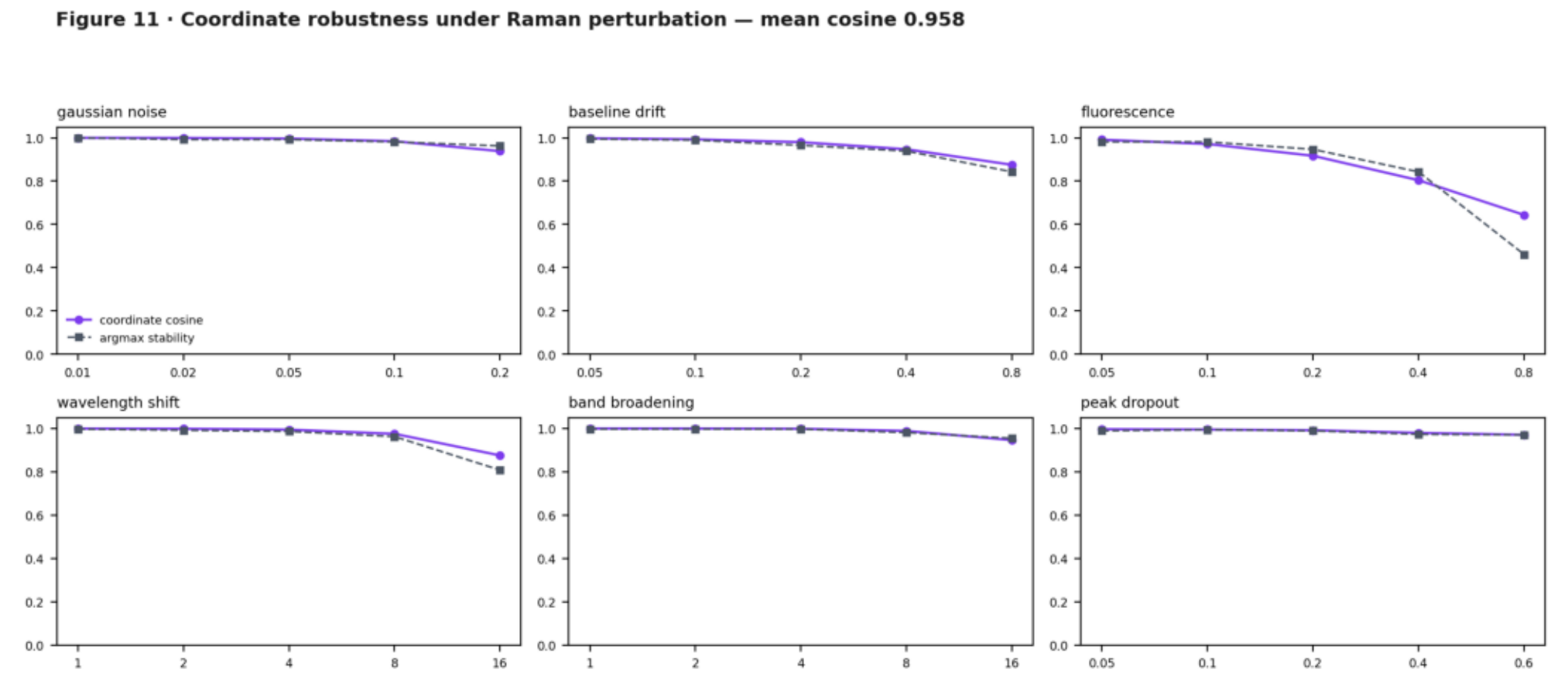
Figure 10

Figure 10 · Continuous Spectral Coordinates — construction and properties



Continuous Spectral Coordinates: kernel x temperature sweep selected on label-free neighbour preservation, and the comparison against hard cluster ids. The coordinates carry nearly double the neighbourhood information of a cluster id.

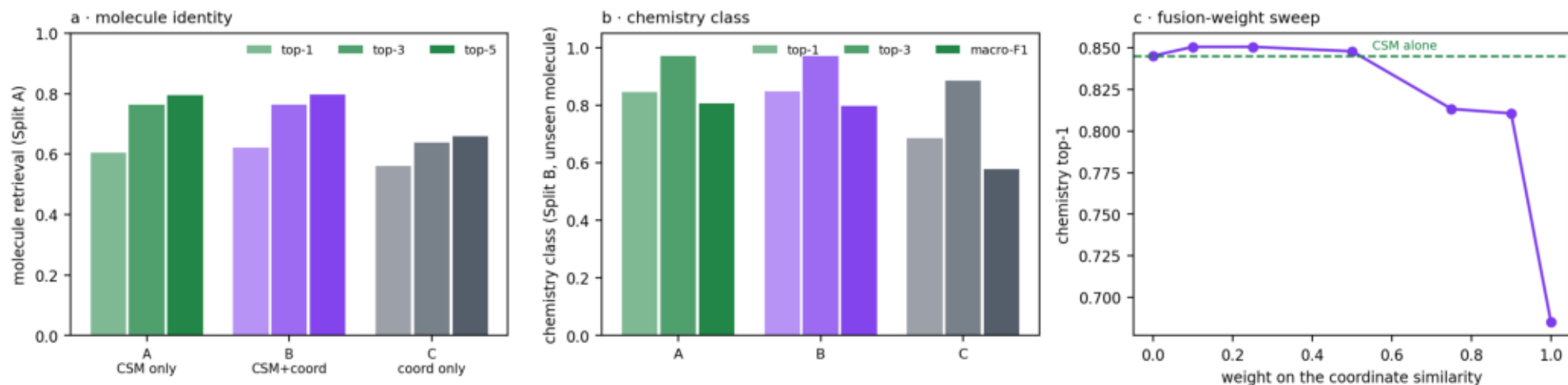
Figure 11



Coordinate robustness under six Raman perturbations at five levels each. Mean coordinate cosine 0.958, argmax stability 0.949.

Figure 12

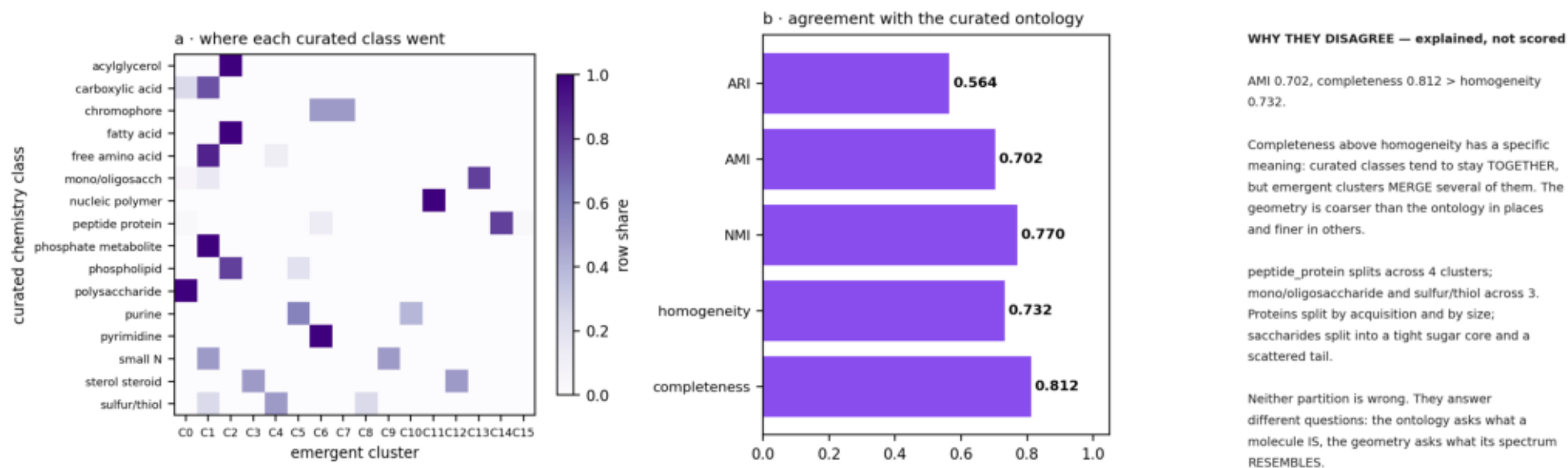
Figure 12 · Retrieval benchmark — molecule-grouped, clustering refitted inside every training fold
A→B molecule $\Delta+0.016$ CI[-0.005,+0.039] McNemar $p=0.180$ · chemistry $\Delta+0.003$ $p=1.000$ — NEITHER IS SIGNIFICANT



The retrieval benchmark, molecule-grouped with the clustering refitted inside every training fold. Adding the coordinate prior changes molecule top-1 by +0.016 with a confidence interval crossing zero ($p = 0.180$) and chemistry top-1 by +0.003 ($p = 1.000$). NEITHER IS SIGNIFICANT. This is the finding the recommendation rests on.

Figure 13

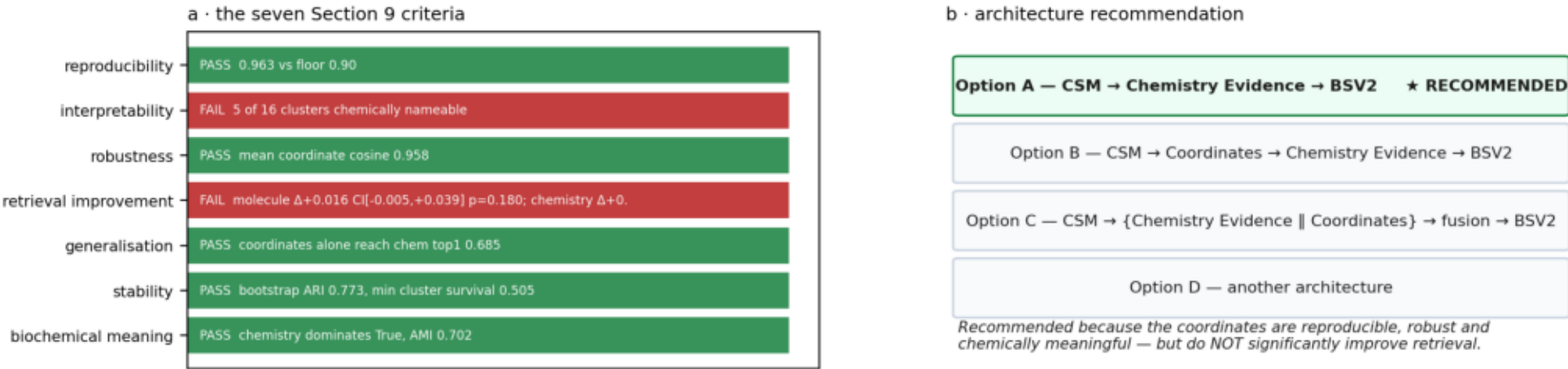
Figure 13 · Emergent geometry versus the curated 16-class ontology



Emergent geometry versus the curated ontology. Completeness (0.812) exceeds homogeneity (0.732): curated classes stay together while emergent clusters merge them. Neither partition is wrong - they answer different questions.

Figure 14

Figure 14 · Decision gate and architecture recommendation



G1 frozen fingerprints verified	G2 no upstream artifact modified	G3 no chemistry label used in geometry construction	G4 Raman-only scope	G5 unit of analysis is the canonical molecule
G6 stability swept over 14 K and 6 algorithms	G6b every internal index computed, none silently NaN	G6c K selection tested for an interior optimum rather than assumed	G7 every cluster classified with a written justification	G8 confounding tested by PERMANOVA with a permutation null
G9 chemistry dominates source and excitation	G10 coordinates non-degenerate	G11 retrieval evaluated molecule-grouped with clustering refitted per fold	G12 no clustering leak across folds	G13 architecture recommendation is evidence-based
G14 Phase 07 not begun				

16 of 16 gates pass · canonical partition average@K=16 (a reporting convention) · audit only, no architecture changed, Phase 07 not begun

Decision gate and architecture recommendation. Five of seven Section 9 criteria pass; the two that fail are the two that would justify an architecture change. Option A is retained and the coordinates remain a scientific instrument.